



"8-Week Build Roadmap"

Document Type:	Implementation Plan
Version:	v1.0
Date:	June 2026
Duration:	8 Weeks
Status:	Ready for Execution

1. Implementation Overview

This implementation plan covers the complete 8-week build cycle for IdeaDNA. Each week is a self-contained sprint with specific deliverables, milestones, and exit criteria. The plan follows agile methodology with daily standups, weekly demos, and continuous integration.

Development Methodology

- Agile Scrum with 1-week sprints
- Daily standups (15 min) — What did I do? What will I do? Any blockers?
- Weekly sprint planning (Monday) + sprint review (Friday)
- Continuous Integration: GitHub Actions for lint, test, build
- Continuous Deployment: Vercel auto-deploy on main branch merge
- Test-Driven Development for critical business logic

Week 1: Foundation & Authentication

Milestone: Project scaffolded, auth working, database connected

Day	Task	Deliverable	Status
Day 1	Initialize Next.js 14 + TypeScript	Project builds, dev server runs	TODO
Day 1	Configure Tailwind + custom theme	Design tokens defined	TODO
Day 2	Setup Prisma + PostgreSQL	Schema defined, migrations run	TODO
Day 2	Seed curated ideas database	500+ ideas in DB	TODO
Day 3	Integrate Clerk Auth	OAuth working, middleware active	TODO
Day 3	Create base layout + navigation	Header, footer, responsive nav	TODO
Day 4	Setup Zustand + React Query	State management working	TODO
Day 4	Configure ESLint + Prettier	Code quality tools active	TODO
Day 5	Setup Vercel deployment	Auto-deploy on push	TODO
Day 5	Week 1 Review + Demo	Working auth + landing page	TODO

Week 2: Idea Generator Core

Milestone: User can input problem and get 5-8 ranked idea cards

Day	Task	Deliverable	Status
Day 6	Build InputParserAgent	Raw input to structured JSON	TODO
Day 6	Build RetrieverAgent	Semantic search in curated DB	TODO
Day 7	Build UpgradeAgent	Personalized idea upgrade	TODO
Day 7	Build ScorerAgent	4-dimension scoring	TODO
Day 8	Create idea cards UI	Hover effects, scores, tags	TODO
Day 8	Implement selection flow	Select triggers lazy-loading	TODO
Day 9	Wire frontend to API	End-to-end flow working	TODO
Day 9	Add loading states	Skeletons, animations	TODO
Day 10	Testing + bug fixes	All flows passing	TODO
Day 10	Week 2 Review + Demo	Live idea generation demo	TODO

Week 3: Evolution Engine

Milestone: Interactive DNA evolution tree with 4-6 mutated versions

Day	Task	Deliverable	Status
Day 11	Build CriticAgent	Weakness analysis working	TODO
Day 11	Build EvolverAgent	Mutation generation	TODO
Day 12	Implement React Flow tree	Interactive nodes + edges	TODO
Day 12	Add node coloring	Score-based colors	TODO
Day 13	Implement recursive evolution	Selected version evolves further	TODO
Day 13	Add lineage tracking	Parent-child DB relations	TODO
Day 14	Create evolution UI page	Input + tree + selection	TODO
Day 14	Wire to API	End-to-end evolution flow	TODO
Day 15	Testing + polish	Tree interactions smooth	TODO
Day 15	Week 3 Review + Demo	Live evolution demo	TODO

Week 4: Code Archaeology

Milestone: 6-agent pipeline analyzing code and reverse-engineering developer thinking

Day	Task	Deliverable	Status
Day 16	Build CodeParserAgent	Language + structure detection	TODO
Day 16	Build GitHub API integration	Repo fetch + file tree	TODO
Day 17	Build PatternDetectorAgent	Design patterns identified	TODO
Day 17	Build IdeaHistorianAgent	Evolution timeline	TODO
Day 18	Build DevPsychologistAgent	Thinking process inference	TODO
Day 18	Build Archaeology report UI	Structured sections	TODO
Day 19	Add 3 input methods	Paste / GitHub URL / Website	TODO
Day 19	Wire to API	Full pipeline working	TODO
Day 20	Testing + polish	All 6 agents accurate	TODO
Day 20	Week 4 Review + Demo	Live archaeology demo	TODO

Week 5: Blueprint Generator

Milestone: 7-section production blueprint with export options

Day	Task	Deliverable	Status
Day 21	Build BlueprintArchitectAgent	7 sections, structured JSON	TODO
Day 21	Create blueprint UI with 7 tabs	Accordion + tabs	TODO
Day 22	Implement Markdown export	Download .md file	TODO
Day 22	Implement PDF export	Styled PDF generation	TODO
Day 23	Add OpenAPI spec export	YAML for Swagger	TODO
Day 23	Add Mermaid diagrams	ER + flow + architecture	TODO
Day 24	Wire lazy-loading trigger	Select -> blueprint generation	TODO
Day 24	Add personalization	Same idea = different blueprint	TODO
Day 25	Testing + polish	All exports valid	TODO
Day 25	Week 5 Review + Demo	Live blueprint demo	TODO

Week 6: UI Mockup Generator

Milestone: Live HTML mockup with device toggle and export

Day	Task	Deliverable	Status
Day 26	Build UIDesignerAgent	HTML + Tailwind generation	TODO
Day 26	Create mockup preview iframe	Live render working	TODO
Day 27	Implement device toggle	Desktop/Tablet/Mobile	TODO
Day 27	Add style variants	5 themes available	TODO
Day 28	Implement HTML export	Download .html	TODO
Day 28	Implement ZIP export	HTML + CSS + assets	TODO
Day 29	Wire lazy-loading trigger	Select -> mockup generation	TODO
Day 29	Add responsive testing	All breakpoints working	TODO
Day 30	Testing + polish	Mockups render correctly	TODO
Day 30	Week 6 Review + Demo	Live mockup demo	TODO

Week 7: Scoring, Profiles, Persistence

Milestone: Radar charts, user onboarding, dashboard, vector search

Day	Task	Deliverable	Status
Day 31	Build D3.js radar chart	4 axes, animated, hover	TODO
Day 31	Create onboarding survey	6 questions, profile saved	TODO
Day 32	Implement context injection	Every agent gets user profile	TODO
Day 32	Build user dashboard	Projects, filters, search	TODO
Day 33	Implement vector search	Similar ideas, similarity > 75%	TODO
Day 33	Add project export (ZIP)	All data packaged	TODO
Day 34	Add shareable links	Public/private toggle	TODO
Day 34	Add evolution lineage view	Tree of parent-child	TODO
Day 35	Testing + polish	Dashboard fully functional	TODO
Day 35	Week 7 Review + Demo	Full platform demo	TODO

Week 8: Polish, Testing, Deploy

Milestone: Production-ready platform deployed and monitored

Day	Task	Deliverable	Status
Day 36	Add Framer Motion animations	Page transitions, hovers	TODO
Day 36	Implement error handling	Toast notifications, fallbacks	TODO
Day 37	Add loading skeletons	Shimmer effects	TODO
Day 37	Implement Redis caching	80% cache hit rate	TODO
Day 38	Add rate limiting	Tier-based, 429 responses	TODO
Day 38	Write E2E tests (Playwright)	Critical flows, 90%+ pass	TODO
Day 39	Performance optimization	Lighthouse > 90, TTFB < 200ms	TODO
Day 39	Security audit	All checklist items passed	TODO
Day 40	Production deployment	Custom domain, SSL, monitoring	TODO
Day 40	Final Review + Launch	Platform live, demo ready	TODO

2. Risks & Mitigations

Risk	Severity	Mitigation
OpenAI API costs exceed budget	HIGH	Implement caching, rate limiting, fallback to cached responses
AI generation latency too high	MEDIUM	Streaming responses, skeleton loaders, progressive enhancement
Database performance degrades	MEDIUM	Connection pooling, read replicas, query optimization
React Flow tree performance	LOW	Virtualization, lazy node rendering, limit tree depth
GitHub API rate limits	MEDIUM	OAuth tokens, request batching, fallback to raw code input
User retention low	HIGH	Onboarding optimization, email reminders, feature announcements

3. Definition of Done

- Code is written, reviewed, and merged to main branch
- Unit tests written with >80% coverage for business logic
- Integration tests passing for API endpoints
- E2E tests passing for critical user flows
- Documentation updated (code comments + README)
- No critical or high-severity bugs open
- Performance benchmarks met (Lighthouse > 90)
- Security checklist passed
- Deployed to staging and verified
- PO/Stakeholder acceptance received

— *End of Implementation Plan* —